

Comparison of Short-Term Estrogenicity Tests for Identification of Hormone-Disrupting Chemicals



The aim of this study was to compare results obtained by eight different short-term assays of estrogenlike actions of chemicals conducted in 10 different laboratories in five countries. Twenty chemicals were selected to represent direct-acting estrogens, compounds with estrogenic metabolites, estrogenic antagonists, and a known cytotoxic agent. Also included in the test panel were 17beta+-estradiol as a positive control and ethanol as solvent control. The test compounds were coded before distribution. Test methods included direct binding to the estrogen receptor (ER), proliferation of MCF-7 cells, transient reporter gene expression in MCF-7 cells, reporter gene expression in yeast strains stably transfected with the human ER and an estrogen-responsive reporter gene, and vitellogenin production in juvenile rainbow trout. 17beta-Estradiol, 17alpha-ethynyl estradiol, and diethylstilbestrol induced a strong estrogenic response in all test systems. Colchicine caused cytotoxicity only. Bisphenol A induced an estrogenic response in all assays. The results obtained for the remaining test compounds--tamoxifen, ICI 182.780, testosterone, bisphenol A dimethacrylate, 4-n-octylphenol, 4-n-nonylphenol, nonylphenol dodecylethoxylate, butylbenzylphthalate, dibutylphthalate, methoxychlor, o,p'-DDT, p,p'-DDE, endosulfan, chlomequat chloride, and ethanol--varied among the assays. The results demonstrate that careful standardization is necessary to obtain a reasonable degree of reproducibility. Also, similar methods vary in their sensitivity to estrogenic compounds. Thus, short-term tests are useful for screening purposes, but the methods must be further validated by additional interlaboratory and interassay comparisons to document the reliability of the methods.